

Problem: The square root of t is greater than 2 and less than 3.5. How many integer values of t satisfy this condition?

SFT (Qwen3-1.7B)

1. Translate the condition into inequalities: $2 < \sqrt{t} < 3.5$
2. Square all parts to eliminate the square root:
 $(2)^2 < t < (3.5)^2$
3. Calculate the squares: $4 < t < 12.25$.
4. Determine the integers strictly between 4 and 12.25: 5, 6, 7, 8, 9, 10, 11, 12.

However, since t must be strictly less than 12.25, 12 is NOT included (it is not strictly less than 12.25).

5. Count: the integers are 5, 6, 7, 8, 9, 10, 11 → 7 values. ✗

+ BC-StepDPO (ours)

1. Translate the condition: $2 < \sqrt{t} < 3.5$
2. Square all parts to eliminate the square root:
 $(2)^2 < t < (3.5)^2$
3. Calculate the squares: $4 < t < 12.25$.
4. Determine the integers with $4 < t < 12.25$: 5, 6, 7, 8, 9, 10, 11, 12.

Since $12 < 12.25$, the value 12 IS included in the range.

5. Count: the integers are 5, 6, 7, 8, 9, 10, 11, 12 → 8 values. ✓

Figure 2. SFT vs. BC-StepDPO: correcting a mathematical reasoning error.